

HW03 逻辑函数化简OK!

1、用代数法求下列逻辑函数的最简与或式:

$$(1) F = (\bar{A}\bar{B} + \bar{A}B + A\bar{B})(\bar{A}C + \bar{B}C + AB)$$

参考解答:

$$\begin{aligned} F &= (\bar{A}\bar{B} + \bar{A}B + A\bar{B})(\bar{A}C + \bar{B}C + AB) \\ &= (\bar{A} + \bar{B})((\bar{A} + \bar{B})C + AB) = \overline{AB}(\overline{ABC} + AB) \\ &= \overline{ABC} = \bar{A}C + \bar{B}C \end{aligned}$$

$$(2) F = AC + \bar{B}C + B\bar{D} + A(B + \bar{C}) + \bar{A}C\bar{D} + A\bar{B}DE$$

参考解答：

$$F = AC + \bar{B}C + B\bar{D} + A(B + \bar{C}) + \bar{A}C\bar{D} + A\bar{B}DE$$

$$= AC + A(B + \bar{C}) + A\bar{B}DE + \bar{A}C\bar{D} + \bar{B}C + B\bar{D}$$

$$= A + \bar{A}C\bar{D} + \bar{B}C + B\bar{D} = A + C\bar{D} + \bar{B}C + B\bar{D}$$

$$= A + \bar{B}C + B\bar{D}$$

$$(3) F = \overline{\overline{A}\overline{C}\overline{D}} + BC + A\overline{C}\overline{D}$$

参考解答:

$$F = \overline{\overline{A}\overline{C}\overline{D}} + BC + A\overline{C}\overline{D} = \overline{\overline{C}\overline{D}} + BC = (C + D)(\overline{B} + \overline{C})$$

$$= \overline{B}C + \overline{B}D + \overline{C}D$$

$$= \overline{B}C + \overline{C}D$$

2、用图解法求下列逻辑函数的最简与或式：

$$(1) F = BD + \bar{A}B\bar{C} + A\bar{C}D + \bar{A}CD + \bar{A}\bar{B}$$

参考解答：

$$F = \bar{A}\bar{B} + \bar{C}D + \bar{A}\bar{C} + BD$$

AB \ CD	00	01	11	10
00	1	1		
01	1	1	1	1
11	1	1	1	
10	1			

$$(2) F(A, B, C, D) = \sum m(0, 1, 2, 4, 5, 8, 9, 10, 12, 13)$$

参考解答:

$$F = \bar{C} + \bar{B}\bar{D}$$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11				
10	1			1

$$(3) F(A, B, C, D) = \prod M(1, 3, 9, 10, 11, 14, 15)$$

参考解答:

$$F = \bar{C}\bar{D} + B\bar{C} + \bar{A}B + \bar{A}\bar{D}$$

AB \ CD	00	01	11	10
00	1	1	1	1
01		1	1	
11		1		
10	1	1		

$$(4) F(A, B, C, D) = \sum m(0, 1, 3, 4, 6, 7, 14, 15) \\ + \sum d(8, 9, 10, 11, 12, 13)$$

参考解答:

$$F = \bar{C}\bar{D} + \bar{B}D + BC \quad \text{or} \quad F = CD + \bar{B}\bar{C} + B\bar{D}$$

AB \ CD	00	01	11	10
00	1	1	d	d
01	1		d	d
11	1	1	1	d
10		1	1	d

AB \ CD	00	01	11	10
00	1	1	d	d
01	1		d	d
11	1	1	1	d
10		1	1	d

3、用图解法求下列逻辑函数的最简与或式和最简或与式：

$$(1) F = \bar{A}\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}CD$$

参考解答：

$$F = \bar{A}B + \bar{B}\bar{C} + A\bar{B}D, \quad F = (\bar{A} + \bar{B})(A + B + \bar{C})(\bar{A} + \bar{C} + D)$$

$$F = (\bar{A} + \bar{B})(A + B + \bar{C})(B + \bar{C} + D)$$

AB \ CD	00	01	11	10
00	1	1	0	1
01	1	1	0	1
11	0	1	0	1
10	0	1	0	0

AB \ CD	00	01	11	10
00	1	1	0	1
01	1	1	0	1
11	0	1	0	1
10	0	1	0	0

AB \ CD	00	01	11	10
00	1	1	0	1
01	1	1	0	1
11	0	1	0	1
10	0	1	0	0

$$(2) F(A, B, C) = \sum m(0, 1, 2, 4, 6)$$

参考解答:

$$F = \bar{A}\bar{B} + \bar{C}, F = (\bar{B} + \bar{C})(\bar{A} + \bar{C})$$

AB \ C	00	01	11	10
0	1	1	1	1
1	1	0	0	0

AB \ C	00	01	11	10
0	1	1	1	1
1	1	0	0	0

$$(3) F(A, B, C, D) = \prod M(5, 7, 13, 15)$$

参考解答:

$$F = \bar{B} + \bar{D}$$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	0	1
11	1	0	0	1
10	1	1	1	1

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	0	1
11	1	0	0	1
10	1	1	1	1

$$(4) F(A, B, C, D) = \sum m(0, 1, 2, 9, 12) + \sum d(4, 6, 10, 11)$$

参考解答:

$$F = \bar{A}\bar{D} + B\bar{C}\bar{D} + \bar{B}\bar{C}D,$$

$$F = (\bar{B} + \bar{D})(\bar{C} + \bar{D})(\bar{B} + \bar{C})(\bar{A} + B + D)$$

$$F = (\bar{B} + \bar{D})(\bar{C} + \bar{D})(\bar{A} + \bar{C})(\bar{A} + B + D)$$

AB \ CD	00	01	11	10
00	1	d	1	0
01	1	0	0	1
11	0	0	0	d
10	1	d	0	d

AB \ CD	00	01	11	10
00	1	d	1	0
01	1	0	0	1
11	0	0	0	d
10	1	d	0	d

AB \ CD	00	01	11	10
00	1	d	1	0
01	1	0	0	1
11	0	0	0	d
10	1	d	0	d